

Expiry-Date Pattern Specification

Reference for the deterministic expiry-date parser · derived from raw RapidOCR output over 58 sample photographs · 03 September 2026

This document catalogues what the expiry information on JustDogs stock actually looks like once RapidOCR has read it, so the parser can be built against real cases rather than assumptions. It has four parts: the **date value formats** (how a date is written), the **label vocabulary** (the words around it, and which mean expiry, manufacture, or a decoy), the **extraction patterns** (the structural cases the parser must resolve), and a **per-image appendix** tagging all 58 photographs.

Scope & honesty note. These patterns were read by hand from the OCR output of every sample. They cover this dataset (9 + 49 images); they are not a closed set for all suppliers JustDogs will ever stock. Treat this as the starting taxonomy, and expect the per-SKU shelf-life table and a review queue to cover what the rules miss.

1. Date value formats

Every way a single date is written on the packs. The parser needs a matcher for each, and a **day-inference rule** for the month-only formats (V3–V5, V9).

ID	Format	Example (as OCR'd)	Seen on	Notes for the matcher
V1	DD/MM/YYYY	20/11/2026, 03-03-2027	many	'/', '-' or '.' separators
V2	DD/MM/YY	22/05/27	38, 48, 49	2-digit year → assume 20YY
V3	MM/YYYY	10/2027	25	day → last day of month
V4	Month YYYY	April 2031, January 2027	2,3,6,9,40,45	full or 3-letter month
V5	MMM-YYYY	FEB-2029, APR-2026	19, 44	hyphen; day → last of month
V6	DD Month YYYY	04 September 2027	34	spelled-out day + month
V7	YYYYMMDD run	20260518	42, 7	8 digits, no separators
V8	YYYY.MMM.DD	2028.JAN.21	39	year-first, dotted
V9	MM/YYYY-MM/YYYY	09/2025-08/2028	26, 29	range; expiry = 2nd half

Day-inference rule (critical). When only month + year are printed (V3, V4, V5, and the second half of V9), the expiry day is the **last day of that month** — confirmed against ground truth: April 2031 → 30/04/31, 10/2028 → 31/10/28. Watch February: FEB-2028 → 29/02/28 (2028 is a leap year). Use a real calendar function, never a hard-coded 28/30/31.

2. Label vocabulary

The words that appear next to a date decide what it means. Three buckets, plus the shelf-life phrases and the indirection markers. Match case-insensitively and tolerate the OCR damage shown — keywords lose spaces, letters and punctuation.

2.1 Expiry labels — the date we want

Canonical	Variants / OCR-damaged forms seen
EXP DATE	EXP, EXP DATE, EXP.DATE, Exp.Date, ERDATE (mangled), EXP :
BEST BEFORE	Best Before, Best Before Date, BestBefore, Best before, BB
EXPIRY DATE	Expiry date, Date of expiry, Expiry Date
USE BEFORE	Use Before, USP (rare mis-read)

2.2 Manufacture labels — used only to derive

Canonical	Variants / OCR-damaged forms seen
MFG DATE	MFG DATE, MFG., MFD, MFD(Date), MF9 (mangled), MFO (mangled)
MANUFACTURING DATE	Manufacturing date, Manufact ring Date (mangled)
MONTH & YEAR OF MFG	Month & Year of Manufacturing, Month and year of manufacture

2.3 Decoy labels — must NOT be picked as expiry

Decoy	Why it is dangerous
Import Date / IMPORT DATE	Looks like a date, often near the expiry (samples 7, 19, 44)
Month & Year of Import	Same; sometimes the only clearly-read date on the panel
Packing / print date	On weighing-scale stickers, an EARLIER date beside the expiry (33/35/37)
Batch / Lot No.	Alphanumeric codes that contain digit runs (2610A250707M046)
MRP / price / per kg	Rs. 680.00 / 272.00 — digit noise on the same sticker

2.4 Shelf-life phrases — trigger derivation

Phrase seen	Meaning
Best Before 1 Year from Mfg. Dt.	expiry = mfg + 12 months (samples 21, 22)
24 months from date of manufacture	expiry = mfg + 24 months
After opening, use within 3 months	relative to OPENING — not a fixed expiry (sample 27)

2.5 Indirection markers — no date in this panel

Marker	Seen on
as printed on bag / as printed on pack	24, 32
MFG. DATE, BEST BEFORE & BATCH NO. SEE BELOW	36
Mfd. By : See On Pack	19, 44

3. Extraction patterns (structural cases)

How the value(s) sit on the panel, and how the parser should resolve each. These are the cases the rule engine branches on. Rough share of the 44 ground-truthed images is given to show where the effort pays off.

E1 Direct labelled expiry

~21 imgs

An expiry label (2.1) immediately followed by a value (section 1). The easy, common case.

EXP DATE:20/11/2026 | Best Before:April 2031 | Expiry date: 15/12/2027

Resolve: Take the value after the label. Apply day-inference if month-only.

E2 Both dates printed, pick expiry

~6

MFG and Best Before both printed as separate labelled values. No maths — just pick the expiry-labelled one and ignore MFG.

Month & Year of Manufacturing: July 2025 / Best Before: January 2027

Resolve: Prefer the expiry label. MFG is there only as a cross-check.

E3 Derive from MFG + shelf life

~4

Only a manufacture date is printed, plus a shelf-life phrase (2.4). Expiry is computed.

MFG DATE 30/07/26 + 'Best Before 1 Year from Mfg. Dt.' → 30/07/27

Resolve: expiry = mfg + N months. Needs a phrase→months map.

E4 Derive, but shelf life NOT on panel

~3

Manufacture date is read, but the shelf-life statement is absent from the OCR text (off-panel or not captured). Cannot be derived from this image alone.

sample 7: MFD.DATE:20250110 only; true expiry 10/01/27 needs '+24 months' known elsewhere

Resolve: Fall back to a per-SKU shelf-life table keyed by barcode.

E5 Compressed MFG-EXP range

~2

Two dates joined into one string, mfg first, expiry second. Often month-year.

09/2025-08/2028 → expiry 31/08/2028 | 11/2025-10/2028 (26)

Resolve: Split on '-', take the later half, apply day-inference.

E6 Weighing-scale / price-gun sticker

~3

In-store printed sticker: a code, an EARLIER packing date, the expiry (later), and a time-of-day. All unlabelled.

51434832CA 23/05/2025 / 23/11/2026 23:42 → expiry 23/11/2026

Resolve: Of the DD/MM/YYYY dates present, the latest is expiry. Strip HH:MM.

E7 Two unlabelled dates

~2

Two bare dates side by side, no labels (or a shared label). The later is expiry, since MFG < EXP.

15/06/26 14/06/27 (36) | 22/05/26 22/05/28 (43)

Resolve: later date wins. This is a heuristic, not a guarantee — low confidence.

E8 Combined label, two values

~1

One label covering both dates: 'Manufacturing / Best before date :'. Values usually mashed together by OCR.

Manufacturing / Best before date : 22/05/26 22/05/28 (43)

Resolve: Split into two dates, later = expiry.

E9 Indirection

~2

The label is present but the value points elsewhere ('as printed on pack', 'SEE BELOW'). No date to extract here.

Best Before ~ as printed on bag (24) | ...SEE BELOW (36)

Resolve: Detect the marker → flag for the other panel / manual entry.

E10 Use-within-N-of-opening

~1

Shelf life is relative to the OPENING event, not manufacture, so no fixed expiry exists.

After opening, use within 3 months (27)

Resolve: Do not compute a date. Store as informational only.

E11 Missing / blank

~4

Labels printed with no values, or nothing date-like read at all.

'Best Before:' and 'Batch No:' printed empty (11, 12); nothing on 8, 28

Resolve: Flag for manual entry; keep image.

E12 Expiry on a different photo

n/a

Barcode/MFG on one face, expiry on another. One photo cannot carry both.

samples 46 + 47 – combine to 04/08/27, neither alone yields it

Resolve: Second-photo path for rigid containers.

4. Suggested parser flow

A first-cut ordering for the deterministic pass. Each step is cheap; the order matters because it resolves the decoys and indirection before trusting a bare date.

1. Normalise OCR text: join lines, fix common damage (spaces in keywords).
2. Detect indirection markers (2.5) → if found and no direct value, flag E9 / manual.
3. Find all EXPIRY-labelled values (2.1) → E1 / E2. Take it, day-infer, done.
4. Else find a RANGE (V9) → E5. Split, later half.
5. Else find a shelf-life phrase (2.4) + a MFG date → E3. Compute.
6. Else collect all bare dates, drop decoys (2.3) and any date ≤ today-ish MFG → later one wins (E6 / E7). Low confidence.
7. Else → E4 / E11: look up shelf life by barcode, or flag for manual entry.

Every derived or heuristic result (E3, E5, E6, E7) is at most medium confidence and must be confirmed by the employee — never auto-saved. Only a directly-labelled, cleanly-parsed date (E1/E2) should be a candidate for auto-save once the pilot proves accuracy. This matches the confidence policy in the architecture proposal.

Sanity rules worth enforcing

- Expiry must be in the future, or only recently past.
- Expiry must be after the manufacture date, when both are read.
- Expiry – MFG should fall within the product's plausible shelf life (e.g. 6–36 months); a 24-month gap on sample 44 is a good cross-check.
- Two captures of the same SKU+batch that disagree → flag, do not overwrite.

5. Appendix — per-image classification

Every sample tagged with its extraction pattern (E-id) and the dominant value format(s). 'Truth' is the ground-truth expiry (DD/MM/YY); blank where none is derivable from a single photo. Use this as the parser's test fixture.

samples1/ (barcode-benchmark set, no ground truth)

Image	Truth	E	Fmt	Dominant OCR evidence
s1/192316	—	E1	V1	EXP DATE:04/12/2026 + MFG
s1/192416	—	E1	V1	ERDATE (label mangled)
s1/192441	—	E1	V1	Expiry date : 10/11/2027
s1/192506	—	E4	V7	MFD+IMPORT YYYYMMDD, no exp
s1/192529	—	E11	—	Best Before blank
s1/192607	—	E1	V1	Best before:18/06/2025
s1/192628	—	E1	V1	Best before:18/06/2025
s1/192642	—	E2	V4	MFD April26 / BB April31
s1/192703	—	E2	V4	Mfg July25 / BB Jan27

samples2/ (expiry-benchmark set, with ground truth)

Image	Truth	E	Fmt	Dominant OCR evidence
s2/1	20/11/26	E1	V1	EXP DATE:20-112026
s2/2	30/04/31	E1	V4	Best Before:April2031
s2/3	30/04/31	E1	V4	BestBefore:April2031
s2/4	30/04/27	E1	V4	BestBefore:April2027
s2/5	12/09/26	E1	V1	EXP DATE:12/09/2026
s2/6	30/04/31	E1	V4	Best Before:April2031
s2/7	10/01/27	E4	V7	MFD 20250110; exp = +24mo off-panel
s2/8	17/02/27	E11	—	nothing date-like read
s2/9	31/01/27	E2	V4	Mfg July25 / BB Jan27
s2/10	03/03/27	E1	V1	Bestbefore:03-03-2027
s2/11	28/02/27	E1	V4	EXP.FEB 2027 (below blank)
s2/12	28/02/27	E11	—	all date labels blank this shot
s2/13	28/02/27	E1	V4	EXP.FEB 2027 (cut-off pack)
s2/14	28/02/27	E1	V4	EXP FEB 2027
s2/15	28/02/27	E1	V4	EXP FEB 2027
s2/16	11/09/26	E11	—	only BATCH read
s2/17	10/03/27	E1	V1	dot-matrix EXP faint on jar

Image	Truth	E	Fmt	Dominant OCR evidence
s2/18	23/07/27	E1	V1	EXP/MFG labels, values missed
s2/19	28/02/29	E1	V5	Expiry Date:FEB-2029 (+Import decoy)
s2/20	13/11/27	E1	V1	EXP:13/11/2027 (poorly read)
s2/21	30/07/27	E3	V1	MFG 30/07/26 + '1 Year from Mfg'
s2/22	28/07/27	E3	V1	MFG 28/07/26 + '1 Years from Mfg'
s2/23	07/03/27	E1	V1	Use Before: 07/03/2027
s2/24	—	E9	—	Best Before ~ as printed on bag
s2/25	31/10/27	E1	V3	Best Before : 10/2027
s2/26	31/10/28	E5	V9	11/2025-10/2028 (range)
s2/27	—	E10	—	use within 3 months of opening
s2/28	31/10/26	E11	—	nothing read
s2/29	31/08/28	E5	V9	09/2025-08/2028 (range)
s2/30	18/02/27	E1	V1	Best Before Date: 18-02-2027
s2/31	15/12/27	E1	V1	Expiry date: 15/12/2027
s2/32	—	E9	—	Best Before: As printed on pack
s2/33	13/08/27	E6	V1	weighing sticker, later date+time
s2/34	04/09/27	E1	V6	Best Before 04 September 2027
s2/35	23/11/26	E6	V1	weighing sticker 23/11/2026 23:42
s2/36	14/06/27	E7	V1	15/06/26 14/06/27 + SEE BELOW
s2/37	28/11/26	E6	V1	weighing sticker
s2/38	22/05/27	E1	V2	MFG28/11/25;EXP22/05/27
s2/39	21/01/28	E1	V8	EXP2028.JAN.21
s2/40	30/04/31	E1	V4	Best Before: April 2031
s2/41	22/06/28	E1	V1	Date of expiry : 22/06/2028
s2/42	18/05/26	E1	V7	EXP20260518
s2/43	22/05/28	E8	V2	combined label, two dates
s2/44	29/02/28	E1	V5	Expiry Date:FEB-2028 (leap!) +decoy
s2/45	30/04/31	E1	V4	Best Before:April 2031
s2/46	—	E12	V2	MFG only; pairs with s2/47
s2/47	—	E12	—	barcode face only
s2/48	23/07/27	E1	V1	EXP :23/07/2027
s2/49	23/07/27	E1	V1	EXP:23/07/2027

Pattern frequency (44 ground-truthed images): E1 direct ≈ 21 · E2 both-printed ≈ 4 · E3 derive ≈ 2 · E4 derive-no-shelf-life ≈ 2 · E5 range ≈ 2 · E6 weighing-sticker ≈ 3 · E7/E8 two-date ≈ 3 · E9 indirection ≈ 2 · E10 open-life ≈ 1 · E11 missing ≈ 4 . Counts are approximate — a few images are OCR-limited rather than pattern-limited.